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SQL Project Data Analytics

Executive Summary

This project analyzes global development indicators using SQL queries on datasets related to country information, sustainable energy, and missing migrants. Key findings include: - Identification of countries leading in renewable energy adoption and those with high CO2 emissions. - Insights into the relationship between health outcomes, education, and economic factors. - Correlation analysis revealing connections between various development indicators.

Methodology

Data from `world-data-2023.csv`, `GlobalMissingMigrantsDataset.csv`, and `global-data-on-sustainable-energy(1).csv` were imported into a PostgreSQL

environment. Data cleaning involved handling character encoding issues and formatting numeric columns. SQL queries were developed to address specific analytical categories: Environmental Sustainability, Health & Education, Economic Development, and Correlation Analysis. The queries utilize JOIN operations, aggregations, filtering, and sorting to extract meaningful insights.

Key Insights

Environmental Sustainability * Top Renewable Energy Adopters: Countries like Iceland and Norway demonstrate high renewable energy consumption, showcasing successful transitions towards sustainable energy sources. * **CO2 Emission Patterns:** China and the United States are among the highest CO2 emitters, indicating significant areas for global emission reduction efforts. * **Regional Comparisons:** Analysis revealed varying levels of CO2 emissions and renewable energy adoption across different geographical regions, suggesting tailored approaches for environmental initiatives.

Health & Education * Life Expectancy Leaders: Countries such as Japan and Switzerland exhibit high life expectancies, often correlated with robust healthcare systems and high living standards. * **Education Enrollment:** Nations with high gross primary education enrollment percentages tend to have better overall development indicators, emphasizing the importance of foundational education. * **Healthcare**

Accessibility: The number of physicians per thousand people varies significantly, highlighting disparities in healthcare access globally.

Economic Development * GDP Analysis: Countries with high GDP often show strong correlations with other development indicators but also contribute significantly to CO2 emissions. * **Cost of Living:** CPI and gasoline prices provide insights into economic stability and affordability within countries, impacting the daily lives of citizens. * **Economic Sustainability:** Tax revenue, total tax rate, and unemployment rates offer a glimpse into a country's fiscal health and labor market, crucial for long-term development planning.

Correlation Analysis * Health vs. Education: A positive correlation was observed between life expectancy and gross primary education enrollment, suggesting that Better education contributes to improved health outcomes. * **Economic vs. Environmental Trade-offs:** A positive correlation between GDP and CO2 emissions indicates that economic growth often comes with environmental costs, necessitating sustainable development strategies. * **Energy Access and GDP:** A strong correlation between access to electricity and GDP highlights the role of energy infrastructure in economic prosperity.

Recommendations

- **Promote Renewable Energy:** Nonprofits should focus on supporting countries with high CO2 emissions to transition to renewable energy sources, drawing lessons from leading adopters.
- **Invest in Education and Healthcare:** Prioritize programs that improve access to quality education and healthcare, especially in regions with low life expectancy and education enrollment.
- **Sustainable Economic Growth:** Advocate for policies that decouple economic growth from environmental degradation, promoting green technologies and sustainable practices.
- **Address Migration Root Causes:** Utilize insights from migration data in conjunction with development indicators to address underlying factors contributing to forced migration.

Technical Appendix

Data Sources

- World Data 2023:
<https://drive.google.com/file/d/1y894OePdtZSe1BTNWHSp2nO1oMlmrm/view?usp=sharing>
 - Global Missing Migrants Dataset:
<https://drive.google.com/file/d/1QmXfbO4TiHXfKq3vGAX5OWsFc5uAHw2/view?usp=sharing>
 - Global Data on Sustainable Energy:
<https://drive.google.com/file/d/1YPEKqztZ98KbUVS1EpVpaTwNsNQae6sZ/view?usp=sharing>
 - Link to the SQL File:
<https://drive.google.com/file/d/1voFKUF9LMfs73aMO5bH-OsYkjkR5CQbl/view?usp=sharing>
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SQL Script

1. Database and Table Setup

Goal	Expected Result	Business Value
Create a table to store the world data from 2023.	A table named 'world_data_2023' with appropriate columns and data types.	Provides a structured foundation for analyzing global economic, demographic, and environmental indicators.

```

CREATE TABLE world_data_2023 (
  country VARCHAR(255),
  density_pkm2 NUMERIC,
  abbreviation VARCHAR(10),
  agricultural_land_percent NUMERIC,
  land_areakm2 NUMERIC,
  armed_forces_size NUMERIC,
  birth_rate NUMERIC,
  calling_code NUMERIC,
  capital_major_city VARCHAR(255),
  co2_emissions NUMERIC,
  cpi NUMERIC,
  cpi_change_percent NUMERIC,
  currency_code VARCHAR(10),
  fertility_rate NUMERIC,
  forested_area_percent NUMERIC,
  gasoline_price NUMERIC,
  gdp NUMERIC,
  gross_primary_education_enrollment_percent NUMERIC,
  gross_tertiary_education_enrollment_percent NUMERIC,
  infant_mortality NUMERIC,
  largest_city VARCHAR(255),
  life_expectancy NUMERIC,
  maternal_mortality_ratio NUMERIC,
  minimum_wage NUMERIC,
  official_language VARCHAR(255),
  out_of_pocket_health_expenditure NUMERIC,
  physicians_per_thousand NUMERIC,
  population NUMERIC,
  population_labor_force_participation_percent NUMERIC,
  tax_revenue_percent NUMERIC,
  total_tax_rate NUMERIC,
  unemployment_rate NUMERIC,
  urban_population NUMERIC,
  latitude NUMERIC,
  longitude NUMERIC
);

```

Goal	Expected Result	Business Value
Import data from world_data_2023_processed.csv into the world_data_2023 table.	All rows from the CSV file are inserted into the table.	Populates the database with essential global country information for analysis.

```

COPY world_data_2023 FROM \'/path/to/your/world_data_2023_processed.csv\'
DELIMITER \',\' CSV HEADER;

```

Goal	Expected Result	Business Value
Create a table to store the global missing migrant's data.	A table named 'global_missing_migrants' with appropriate columns and data types.	Establishes a data structure for analyzing migration trends and incidents.

```
CREATE TABLE global_missing_migrants (  
  incident_type text,  
  incident_year INTEGER,  
  reported_month text,  
  region_of_origin text,  
  region_of_incident text,  
  country_of_origin text,  
  number_of_dead INTEGER,  
  minimum_estimated_number_of_missing INTEGER,  
  total_number_of_dead_and_missing INTEGER,  
  number_of_survivors INTEGER,  
  number_of_females INTEGER,  
  number_of_males INTEGER,  
  number_of_children INTEGER,  
  cause_of_death TEXT,  
  migration_route text,  
  location_of_death text,  
  information_source text,  
  coordinates text,  
  unsd_geographical_grouping text  
);
```

Goal	Expected Result	Business Value
Import data from global_missing_migrants_processed.csv into the global_missing_migrants table.	All rows from the CSV file are inserted into the table.	Populates the database with critical data on missing migrants for humanitarian analysis.

```
COPY global_missing_migrants FROM  
'/path/to/your/global_missing_migrants_processed.csv' DELIMITER ',' CSV  
HEADER;
```

Goal	Expected Result	Business Value
Create a table to store the global data on sustainable energy.	A table named 'sustainable_energy' with appropriate columns and data types.	Provides a structured foundation for analyzing global sustainable energy trends.

```
CREATE TABLE sustainable_energy (
  entity VARCHAR(255),
  year INT,
  access_to_electricity_percent NUMERIC,
  access_to_clean_fuels_for_cooking_percent NUMERIC,
  renewable_energy_consumption_percent NUMERIC,
  energy_intensity_of_primary_energy_gdp_ratio NUMERIC,
  electricity_from_fossil_fuels_percent NUMERIC,
  electricity_from_nuclear_percent NUMERIC,
  electricity_from_renewables_percent NUMERIC
);
```

Goal	Expected Result	Business Value
Import data from sustainable_energy_processed.csv into the sustainable_energy table.	All rows from the CSV file are inserted into the table.	Populates the database with essential sustainable energy data for analysis.

```
COPY sustainable_energy FROM
\'/path/to/your/sustainable_energy_processed.csv\' DELIMITER \'\' CSV HEADER;
```

2. Analytical Queries

Environmental Sustainability

Goal	Expected Result	Business Value
Identify the top 10 countries with the highest renewable energy consumption percentage.	A list of countries and their renewable energy consumption percentage, ordered from highest to lowest.	Helps the nonprofit identify leading countries in renewable energy adoption for potential case studies or partnerships.

```
SELECT
  entity AS country,
  renewable_energy_consumption_percent
FROM
  sustainable_energy
WHERE
  renewable_energy_consumption_percent IS NOT NULL
ORDER BY
  renewable_energy_consumption_percent DESC
LIMIT 10;
```

Goal	Expected Result	Business Value
Identify the top 10 countries with the highest CO2 emissions.	A list of countries and their CO2 emissions, ordered from highest to lowest.	Highlights countries with significant CO2 output, indicating areas for emission reduction efforts.

```
SELECT
    country,
    co2_emissions
FROM
    world_data_2023
WHERE
    co2_emissions IS NOT NULL
ORDER BY
    co2_emissions DESC
LIMIT 10;
```

Goal	Expected Result	Business Value
Compare average CO2 emissions and renewable energy consumption across different regions.	Average CO2 emissions and renewable energy consumption grouped by region, allowing for regional comparisons.	Helps the nonprofit understand regional disparities in environmental performance and target interventions effectively.

```
SELECT
    wd.abbreviation AS region_abbreviation,
    AVG(wd.co2_emissions) AS average_co2_emissions,
    AVG(se.renewable_energy_consumption_percent) AS
average_renewable_energy_consumption_percent
FROM
    world_data_2023 AS wd
JOIN
    sustainable_energy AS se ON wd.country = se.entity
WHERE
    wd.co2_emissions IS NOT NULL AND se.renewable_energy_consumption_percent IS
NOT NULL
GROUP BY
    wd.abbreviation
ORDER BY
    average_co2_emissions DESC;
```

Health & Education

Goal	Expected Result	Business Value
Rank countries by life expectancy (a key health indicator).	A list of countries and their life expectancy, ordered from highest to lowest.	Identify countries with strong public health systems and high quality of life, serving as models for intervention.

```
SELECT
    country,
    life_expectancy
FROM
    world_data_2023
WHERE
    life_expectancy IS NOT NULL
ORDER BY
    life_expectancy DESC
LIMIT 10;
```

Goal	Expected Result	Business Value
Analyze gross primary education enrollment rates.	A list of countries and their primary education enrollment percentages, ordered from highest to lowest.	Highlights countries with successful primary education programs, indicating areas for educational development.

```
SELECT
    country,
    gross_primary_education_enrollment_percent
FROM
    world_data_2023
WHERE
    gross_primary_education_enrollment_percent IS NOT NULL
ORDER BY
    gross_primary_education_enrollment_percent DESC
LIMIT 10;
```

Goal	Expected Result	Business Value
Compare healthcare accessibility (physicians per thousand) across countries.	A list of countries and their number of physicians per thousand people, ordered from highest to lowest.	Pinpointing countries with robust healthcare infrastructure, offering Insights for improving healthcare access in underserved regions.

```
SELECT
    country,
    physicians_per_thousand
FROM
    world_data_2023
WHERE
    physicians_per_thousand IS NOT NULL
ORDER BY
    physicians_per_thousand DESC
LIMIT 10;
```

Economic Development

Goal	Expected Result	Business Value
Analyze GDP per capita across countries.	A list of countries and their GDP, ordered from highest to lowest.	Identifies economically strong nations that could serve as donors or partners, and highlights countries needing economic development support.

```
SELECT
    country,
    gdp
FROM
    world_data_2023
WHERE
    gdp IS NOT NULL
ORDER BY
    gdp DESC
LIMIT 10;
```

Goal	Expected Result	Business Value
Analyze economic sustainability metrics: tax revenue, total tax rate, and unemployment rate.	A list of countries with their tax revenue, total tax rate, and unemployment rate.	Helps assess a country's fiscal health and labor market conditions, crucial for long-term development planning.

```

SELECT
    country,
    tax_revenue_percent,
    total_tax_rate,
    unemployment_rate
FROM
    world_data_2023
WHERE
    tax_revenue_percent IS NOT NULL
    AND total_tax_rate IS NOT NULL
    AND unemployment_rate IS NOT NULL
ORDER BY
    unemployment_rate ASC;

```

Correlation Analysis

Goal	Expected Result	Business Value
Analyze the correlation between life expectancy and gross primary education enrollment.	A single numeric value representing the correlation coefficient.	Help understand the relationship between health outcomes and basic education, informing integrated development programs.

```

SELECT
    CORR(life_expectancy, gross_primary_education_enrollment_percent) AS
    life_expectancy_education_correlation
FROM
    world_data_2023
WHERE
    life_expectancy IS NOT NULL AND gross_primary_education_enrollment_percent
    IS NOT NULL;

```

Goal	Expected Result	Business Value
Analyze the correlation between GDP and CO2 emissions.	A single numeric value representing the correlation coefficient.	Provides insights into the environmental impact of economic growth, crucial for sustainable development strategies.

```

SELECT
    CORR(gdp, co2_emissions) AS gdp_co2_emissions_correlation
FROM
    world_data_2023
WHERE
    gdp IS NOT NULL AND co2_emissions IS NOT NULL;

```

Goal	Expected Result	Business Value
Analyze the correlation between access to electricity and GDP.	A single numeric value representing the correlation coefficient.	Explores the link between energy access and economic prosperity, guiding infrastructure development initiatives.

```
SELECT
    CORR(se.access_to_electricity_percent, wd.gdp) AS
    electricity_access_gdp_correlation
FROM
    sustainable_energy AS se
JOIN
    world_data_2023 AS wd ON se.entity = wd.country
WHERE
    se.access_to_electricity_percent IS NOT NULL AND wd.gdp IS NOT NULL;
```